# Experimental grid, checkpoints, and compute budget

> **Reading note.** This is an explanatory reconstruction of Jiaxuan Zou's literature synthesis. Published experiments belong to the cited teams. Numbered protocols and illustrative matrices are our planning synthesis unless explicitly attributed.

The blog's §5 explains why the geometry of the experiments must match the question. If every run uses $D = qN$ for a constant token-to-parameter ratio $q$, size and data are perfectly correlated. A fit cannot reliably identify their separate effects from that line alone.

## Choose a layout

| Layout | Useful for | Main risk |
|---|---|---|
| Cartesian $(N, D)$ grid | Separate size and token effects | Expensive tuning |
| IsoFLOP slices | Compute-optimal allocation | Missing minima if sizes span too narrow a range |
| L-shaped sampling | Economical surface estimation | Interior predictions depend strongly on functional form |
| Fixed-TPP progression | Forecast a prescribed recipe | Cannot identify a general two-dimensional surface |

An illustrative grid uses three fitting sizes and three token horizons per size, plus a fourth size reserved for holdout. Choose numerical sizes from measured throughput and target TPP, not from the labels of a published model family. For an IsoFLOP budget $C_i$, the approximation gives $D_{ij} = C_i/(6N_j)$; substitute actual FLOP accounting before launching.

## Procedure

1. Pilot the smallest models to measure noise, stability, and throughput.
2. Reserve tuning, replication, annealing, failure-recovery, and holdout compute.
3. Spread fitting points across both axes needed for the target decision.
4. Reserve whole size or compute tiers to test the actual extrapolation direction.
5. Evaluate trajectories at comparable progress fractions and decide how early transients will be excluded before seeing final holdouts.
6. Use remaining budget to resolve the dominant uncertainty: seeds for noise, larger tiers for extrapolation, or high-TPP runs for horizon mismatch.

Checkpoints share initialization and data history. WSD branches share their prefix. Count prefix compute once in the budget and cluster related observations together when estimating uncertainty.

**Deliverable:** a run table with size, tokens, FLOPs, seed, branch parent, tuning budget, fit/development/holdout role, and contingency allocation.

**Sources:** blog §5; Hoffmann et al., 2022, Chinchilla; Choshen et al., 2025; Skaling, 2026.

Blog/notes crosswalk · Detailed companion